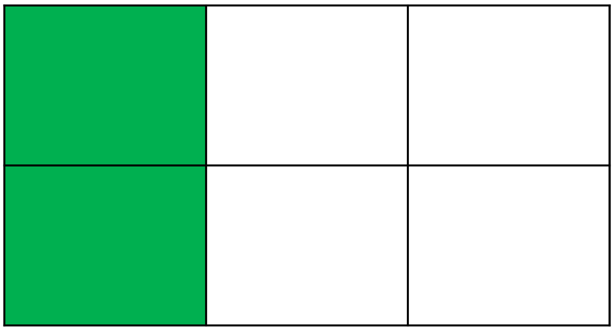
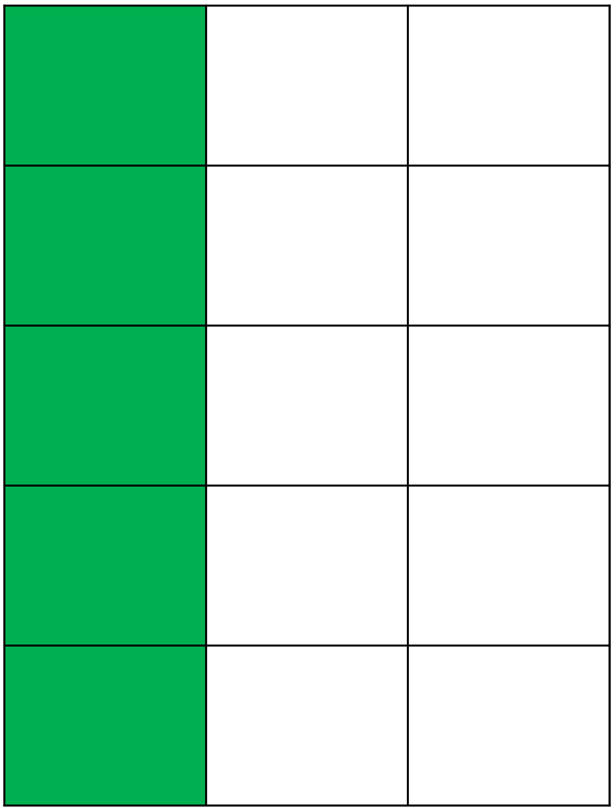
Competence	Awareness
A. Knowledge and Understanding	
A1 - Have maintained and extended a sound theoretical approach to enable them to develop their particular role.	
A2 - Are developing technological solutions to unusual or challenging problems, using their knowledge and understanding and/or dealing with complex technical issues or situations with significant levels of risk.	
B. Design, Development and Solving Engineering Problems	
B1 - Take an active role in the identification and definition of project requirements, problems and opportunities	
B2 - Can identify the appropriate investigations and research needed to undertake the design, development and analysis required to complete an engineering task and conduct these activities effectively	
B3 - Can implement engineering tasks and evaluate the effectiveness of engineering solutions	
C. Responsibility, Management and Leadership	
C1 - Plan the work and resources needed to enable effective implementation of a significant engineering task or project	
C2 - Manage (organise, direct and control), programme or schedule, budget and resource elements of a significant engineering task or project	
C3 - Lead teams or technical specialisms and assist others to meet changing technical and managerial needs	
C4 - Bring about continuous quality improvement and promote best practice	
D. Communication and Interpersonal Skills	
D1 - Communicate effectively with others, at all levels, in English	

D2 - Clearly present and discuss proposals, justifications and conclusions	
D3 - Demonstrate personal and social skills and awareness of diversity and inclusion issues.	
E. Personal and Professional Commitment	
E1 - Understand and comply with relevant codes of conduct	
E2 - Understand the safety implications of their role and manage, apply and improve safe systems of work	
E3 - Understand the principles of sustainable development and apply them in their work	
E4 - Carry out and record the Continuing Professional Development (CPD) necessary to maintain and enhance competence in their own area of practice	
E5 - Understand the ethical issues that may arise in their role and carry out their responsibilities in an ethical manner.	

Working	Practitioner	Expert





Evidence
Applied and extended theoretical knowledge across diverse projects at Heriot-Watt (imitation learning, evolutionary robotics, machine vision, embedded systems, PSO-ANN) and at UoM (mobile manipulation, ROS 2, MoveIt, Nav2).
Developed solutions to challenging engineering problems across multiple projects, including imitation learning for real-time robotic tracking, behaviour-based and evolutionary rover navigation, and an autonomous mobile manipulation system.
Identified and defined project requirements across group and individual projects at both Heriot-Watt and UoM, including defining manipulation and mechanical requirements for the Leo Rover mobile manipulation system.
Conducted independent research across multiple projects to select appropriate methods - including comparing YOLO, traditional CV, and HSV segmentation for object detection, and evaluating motion planning approaches for robotic manipulation.
Implemented and evaluated engineering solutions across multiple projects, including the manipulation subsystem and mechanical sled design for the Leo Rover, the imitation learning pipeline for the Panda arm, and the PSO-ANN classifier.
Communicated technical work through academic reports across multiple modules, presentations and meetings. As well as a personal portfolio website documenting major projects.

Presented and justified design decisions through project reports and a conference publication, clearly articulating methodology, trade-offs, and evaluation results.
Worked in multicultural project teams across both MEng and MSc, including a 5-perso

Consistently adhered to academic integrity, professional conduct, and safety regulatio
Applied safety-conscious design across multiple projects, including safe manipulation constraints in the Leo Rover system (joint limits, collision checking, E-stop) and fail-safe behaviour in embedded robotics projects.
Designed the ropeless fishing gear demonstrator as a sustainable alternative to rope-based fishing, directly addressing marine entanglement risk - a recognised ecological concern.
Currently completing a structured CPD reflection and personal development plan
Considered ethical implications in projects involving autonomous systems and medical applications, including responsible use of AI in clinical contexts and safe autonomous decision-making.

›n international team for the Leo Rover project.

›ns throughout MEng and MSc project work.